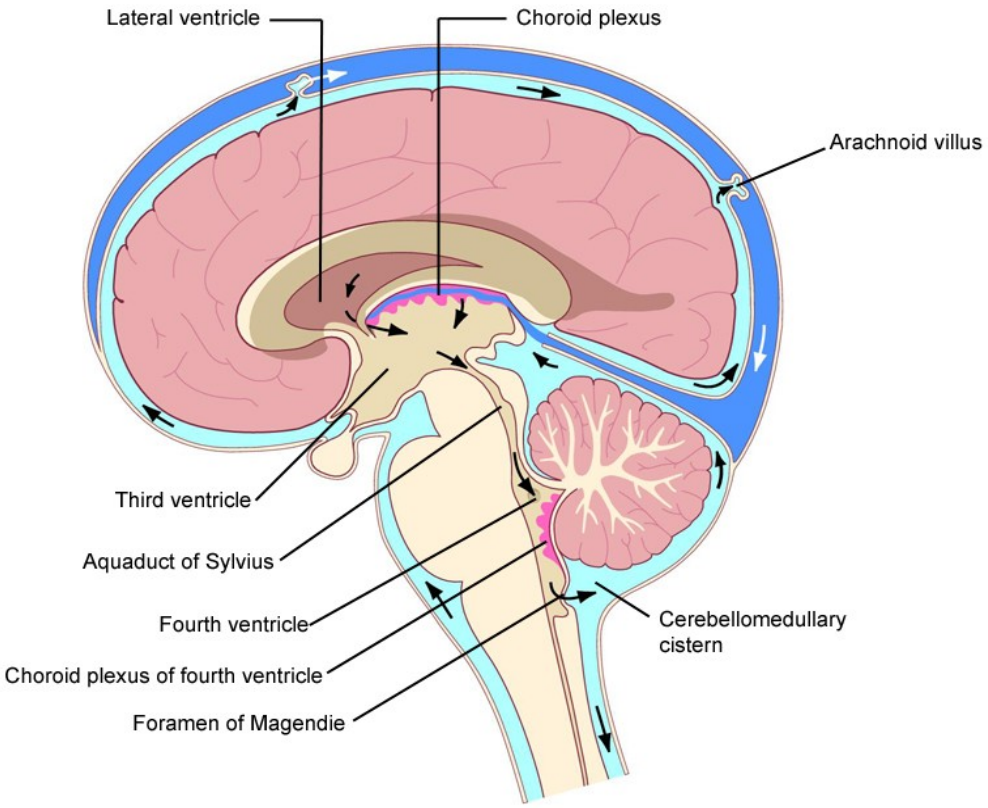
Cerebrospinal fluid (CSF) is the fluid that surrounds the brain and spinal cord and serves primarily as a shock absorber to protect vital structures from trauma. It is found between the pia mater and arachnoid mater (the subarachnoid space).

Normal CSF production in adults is about 21 mL/hour or about half a liter a day. The total volume of CSF at any given time, however, is only about 150 mL. CSF is formed primarily by the choroid plexuses in the ventricles. It flows from the lateral ventricles to the third ventricles via the foramen of Monro, then passes into the fourth ventricle via the aqueduct of Sylvius. It then passes from the fourth ventricle to the cerebellomedullary cistern via the foramen of Magendie. From the cerebellomedullary cistern, it enters the subarachnoid space, circulates around the brain and spinal cord, and then is reabsorbed in the arachnoid villi above the cerebral hemispheres.




In order to produce CSF, the choroid plexuses actively secrete sodium to produce a fluid that is isotonic despite the fact that it contains a lower potassium, bicarbonate, and glucose concentration than plasma. CSF also contains trace amounts of protein.

Factors that decrease the production of CSF include carbonic anhydrase inhibitors, corticosteroids, isoflurane, spironolactone, furosemide, and vasoconstrictors.

*Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579-580.*

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